PALAK SHARMA

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SUMMARY

Final-year B.Tech (Software Engineering) student with experience in Python and AI/ML projects. Winner of SIH 2025 for building an AI-based DCRM system for fault detection in EHV circuit breakers. Strong problem-solving skills, passionate about scalable solutions and real-world ML applications.

WORK EXPERIENCE

Machine Learning Trainee — **jan 2024 - march 2024**

- Gained hands-on experience in Python, Numpy, and Pandas for data analysis and model building .
- Performed data preprocessing, handled missing values, and applied feature selection techniques to improve model performance.
- Contributed to project enhancements within a team environment, utilizing Git & GitHub for basic version control and code sharing.

SKILLS & INTERESTS

- **Programming Languages:** Python
- **Frameworks & Tech**: Numpy, Pandas, Flask
- **Database Management**: MySQL
- **Software Concepts**: Object-Oriented Programming (OOPs)
- **Tools & Platforms**: Git & GitHub, Visual Studio Code (VS Code)

SOFT SKILLS

- Problem Solving, Communications, Time Management, Teamwork

EDUCATION

Bachelor of Technology in Computer Science and Information Technology — **2022-2026**

Institute of Engineering & Science, IPS Academy. Current CGPA - 7.8

PROJECT

SkyPredict: Explainable Flight Delay Monitoring Platform)

- **Key Tech:** Python, Flask, Scikit-learn, Pandas
- Developed a flight delay prediction system using Random Forest with probability-based risk.
- Implemented Explainable AI (XAI) to identify key contributing factors affecting delays.
- Designed and deployed a Flask-based web application with an interactive UI.
- Built an end-to-end pipeline including data preprocessing, model training, and inference.

VOLTGUARD – AI (DCRM Analysis System for EHV Circuit Breakers)

- **Key Tech**: Python, Machine Learning, Data Analysis
- Built an AI-based system to analyze DCRM (CSV) test data for fault detection.
- Visualized data by converting raw signals into meaningful graphs.
- Compared results with healthy reference signatures for anomaly detection.
- Developed an ML model to calculate deviations (deltas) and predict root causes of faults.

ACHIEVEMENTS

- Smart India Hackathon (SIH2025) - Winner organised by Ministry of Power (MOP), Dec 2025
- BSF Event (2024) - Cultural Performer, Fed 2024

CERTIFICATIONS

- Oracle Certified in AI Foundation, Nov 2025
- TATA -GenAI Powered Data Analytics Job Simulation,March 2026